Context-Tree Models as Character-Level Language Models: Empirical Evaluation and the Letter-vs-Block Granularity Problem

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Abstract

I present an empirical evaluation of context-tree language models inspired by Context-Tree Weighting (CTW) (Willems et al., 1995) on the WikiText-2 benchmark. The implemented large-alphabet model, TextCTW, interpolates KT estimates along the observed context path; it is not the exact multinomial CTW mixture. TextCTW with depth $D = 7$ achieves 2.162 bits per character (BPC), compared to 1.154 BPC for GPT-2 small — a gap of approximately 1.008 BPC, or roughly a factor of two in per-character information. To locate the origin of this gap, I compute the empirical conditional entropy $\widehat{H}(D) = \widehat{H}(X_t \mid X_{t-D:t-1})$ as an information-theoretic lower bound on any D-memory model. At $D = 7$, $\widehat{H}(7) = 1.205$ BPC, placing the fundamental class gap (finite-D vs. long-range) at only 0.051 BPC; the dominant contribution of 0.957 BPC is a data-scarcity gap attributable to tree sparsity past the theoretical saturation depth $D_{\max} \approx 4$. Per-word-position analysis reveals that virtually all of the excess loss is incurred at the first character of each word (+2.628 BPC vs. GPT-2), precisely where GPT-2's long-range context provides word-identity information. I also evaluate word-granularity CTW variants across vocabulary sizes and depths, confirming that the depth-one model is always optimal for all tested vocabulary sizes due to $D_{\max} < 2$. These findings connect to the theoretical framework of Willems (2026) (Section 8), which formalises when block-level (token) CTW beats letter-level CTW under matched memory, and motivates the conjecture that an optimal block length L^* exists.

1 Introduction

The Context-Tree Weighting (CTW) algorithm, introduced by Willems, Shtarkov, and Tjalkens (Willems et al., 1995), provides a principled Bayesian approach to sequential data compression for sources with bounded memory. Given a context depth D, CTW computes the exact Bayesian marginal likelihood over all context trees of depth at most D in $O(D)$ time per symbol — with no model selection and no pruning. For a binary Markov chain of order $m \le D$, CTW achieves the optimal KT redundancy of $\frac{d}{2} \log n$ bits, where $d = a^m(a-1)$ is the number of free parameters (Rissanen, 1984).

In his Shannon Lecture at ISIT 2026 (Willems, 2026), Frans Willems raised the question of how CTW performs as a language model on natural text, both at the character level and at the token level. The lecture notes that letter-level CTW achieves competitive compression on literary texts (Poe, Austen, Twain) and that the extracted context trees reveal genuine linguistic structure (common suffixes as contexts). More importantly, the lecture conjectures that *token-level* CTW can outperform letter-level CTW in overall code length, and poses the theoretical question of when this advantage exists and how large it is — the *letter-vs-block granularity problem* (Section 8 of the lecture notes).

The present work addresses three complementary questions:

1. How large is the gap between the best finite-memory model (CTW at depth D) and a modern neural language model (GPT-2) on a standard benchmark, and where does the loss originate?

2. What is the information-theoretic floor for any D-memory model, and does the CTW gap arise from the wrong model class (insufficient memory) or from data scarcity?

3. How do word-granularity CTW models perform relative to letter-level CTW, and what does this say about the letter-vs-block conjecture of Willems (2026)?

Main findings.

1. TextCTW $D = 7$ achieves 2.162 BPC; GPT-2 small achieves 1.154 BPC.

2. The information-theoretic lower bound $\widehat{H}(7) = 1.205$ BPC shows that the *class gap* is only 0.051 BPC: a perfect D-memory model would nearly close the gap with GPT-2. The dominant 0.957 BPC is a *data-scarcity* gap caused by tree sparsity past $D_{\max} \approx 4$.

3. Virtually all excess loss (+2.628 BPC over GPT-2) concentrates at word-initial positions; within words, TextCTW is within 0.6 BPC of GPT-2.

4. Word-CTW with depth $D = 1$ achieves 1.774 BPC (excluding space cost); larger depth always hurts because $D_{\max} < 2$ for all tested vocabularies.

5. The node-growth experiment confirms the $D_{\max}$ saturation theory: empirical context counts drop below 1% of theoretical capacity at depth $D = D_{\max} + 1$.

The paper is organised as follows. Section 2 reviews CTW and the theoretical framework from Willems (2026). Section 3 describes the experimental setup. Sections 4–8 present results. Section 9 connects empirical findings to the theory. Section 10 concludes with open problems.

2 Background

2.1 CTW Algorithm

Let $x_1, x_2, \ldots, x_n$ be a sequence over a finite alphabet A with $|A| = a$. A *context tree* of depth D is an a-ary tree whose leaves define the prediction contexts. CTW maintains a full D-level tree and, at each node s, a weighted probability

$$P_w^{(s)} \;=\; \tfrac{1}{2}\, P_{\mathrm{KT}}^{(s)} \;+\; \tfrac{1}{2} \prod_{\text{children } c} P_w^{(c)}, \tag{1}$$

where $P_{\mathrm{KT}}^{(s)}$ is the Krichevsky–Trofimov (KT) estimate at node s (Willems et al., 1995). The KT estimate for observing symbol b given context s is

$$P_{\mathrm{KT}}(b \mid s) \;=\; \frac{n_b(s) + \alpha}{\sum_{b'} n_{b'}(s) + a\alpha}, \tag{2}$$

where $n_b(s)$ is the count of symbol b after context s and $\alpha = 1/2$ is the standard Dirichlet pseudocount. The root node probability $P_w^{(\varepsilon)}$ is the exact Bayesian marginal likelihood under a prior that assigns weight $2^{-(2^{d+1}-1)}$ to each tree of depth d.

For large alphabets (text), the multinomial KT estimator generalises to Dirichlet$(1/2, \ldots, 1/2)$. I use $\alpha = 0.5$ for character-level models and $\alpha = 0.01$ for word-level models (to handle large vocabularies without pseudocount inflation).

2.2 Metrics

Bits per character (BPC). All results are reported in BPC, defined as BPC $= -\frac{1}{n}\sum_{t=1}^{n}\log_2 P(x_t \mid x_{1:t-1})$. Lower BPC is better.

$D_{\max}$ **— maximum useful depth.** For a corpus of N tokens over alphabet of size a, and a minimum-count threshold $n_{\min}$, the maximum depth at which the context tree is statistically meaningful is

$$D_{\max}(a, N, n_{\min}) \;=\; \frac{\log(N/n_{\min})}{\log a}. \tag{3}$$

At depth $D > D_{\max}$, most contexts have fewer than $n_{\min}$ observations, so the KT estimate reverts to the prior and contributes no useful signal.

$\widehat{H}(D)$ **— empirical conditional entropy.** The empirical D-th order conditional entropy

$$\widehat{H}(D) \;=\; \widehat{H}(X_t \mid X_{t-D:t-1}) \;=\; -\frac{1}{n}\sum_{(c,x)} n(c,x)\,\log_2\frac{n(c,x)}{n(c)} \tag{4}$$

is the lower bound on BPC achievable by *any* D-memory model trained on the same data. Here c ranges over all observed D-grams and x over the alphabet.

2.3 The Letter-vs-Block Granularity Problem

The following theoretical problem is posed in Willems (2026) (Section 8).

Setup. Source: stationary Markov chain of order m on alphabet A, $|A| = a$, with transition probabilities bounded away from zero.

- **Letter CTW:** context depth D, alphabet size a.
- **Block CTW:** block length L, context depth D_b, alphabet size a^L (one token $= L$ letters).
- **Matched memory:** $D = L \cdot D_b$ (both models see at most D past letters).

Quantity. The total redundancy of assignment Q is

$$R_n(Q) \;=\; \sup_{\theta} D\big(P_\theta^{X^n} \,\big\|\, Q^{X^n}\big) + \frac{\Gamma}{2}\log n, \tag{5}$$

where Γ is the number of free leaf parameters. The key asymptotic comparison is the limit

$$\lim_{n\to\infty}\frac{R_n^{\text{letter}} - R_{n/L}^{\text{block}}}{\log n}. \tag{6}$$

Conjecture 1 (Willems 2026, Conjecture 8.4)**.** *There exist thresholds $L_*(a, m, \varepsilon)$ and $n_*(a, m, L)$ such that:*

1. *If $m \le D$ and L is small, then for n large enough:*

$$R_{n/L}^{\text{block}} + \tfrac{(a^L-1)\Gamma_b}{2}\log\tfrac{n}{L} \;<\; R_n^{\text{letter}} + \tfrac{(a-1)\Gamma_\ell}{2}\log n.$$

Block CTW wins because shorter contexts suffice to capture order-m dependencies.

2. *If L is too large (a^L grows), the block alphabet's $\frac{a^L}{2}\log n$ parameter cost dominates and letter CTW wins.*

So there is an optimal granularity L^* minimising total redundancy under fixed letter-memory D.

Minimal first theorem (Section 8.5). For binary sources ($a = 2$), Markov order $m = 1$, matched depth $D = 2L$, the target is to give the exact asymptotic of (6) as a function of L; its sign determines which model class wins.

My experiments on word-level CTW provide direct empirical evidence for this conjecture.

3 Experimental Setup

Dataset. I use WikiText-2, a benchmark derived from verified Good and Featured Wikipedia articles. The training split (2.09M tokens) provides 10,023,770 characters after normalisation ($\approx$ 1.69M words). The validation split provides 1,051,557 characters.

Normalisation. All text is lowercased, punctuation is collapsed to a single `` `.' `` symbol, and digits are replaced by `` `0' ``. The resulting alphabet has $a = 28$ characters (26 letters + space + punctuation/digit).

Models.

- **TextCTW** (this work): an interpolated large-alphabet context-tree predictor implemented from scratch. Each node uses the multinomial KT estimator, and predictions are mixed along the single observed context path; depths $D \in \{3, 5, 7, 10\}$.
- **KT n-gram baseline**: suffix-backoff KT n-gram with context depths $D \in \{3, 5, 7, 10\}$.
- **GPT-2 small** (117M parameters): pretrained; evaluated with 512-token sliding window.
- **GPT-2 small (fine-tuned)**: 3 epochs on WikiText-2 training split, learning rate 5×10^{-5}, block size 512.

Implementation scope. Exact CTW is implemented separately for binary alphabets. The character- and word-level experiments use TextCTW, a practical single-path interpolation for large alphabets rather than the full multinomial CTW product over all child subtrees. For continuity with the experiment scripts and figure legends, some tables abbreviate TextCTW as "CTW"; all reported large-alphabet results refer to this approximation.

Cross-corpus control. I additionally evaluate on text8 (27-character Wikipedia excerpt, 2M train / 200K val chars) to confirm robustness of results.

4 Character-Level TextCTW Results

4.1 Depth Sweep

Table 1 reports BPC for TextCTW and the KT n-gram baseline on the WikiText-2 validation set.

Table 1: Character-level BPC on WikiText-2 validation (1M chars). $N_{\text{train}} = 10{,}023{,}770$ chars; $|A| = 28$.

Model	**Depth** D	**BPC**	**Perplexity**
TextCTW	3	2.557	5.885
TextCTW	5	2.170	4.500
TextCTW	**7**	**2.162**	**4.476**
TextCTW	10	2.184	4.544
KT n-gram (backoff)	3	2.371	5.173
KT n-gram (backoff)	**5**	**1.985**	**3.959**
KT n-gram (backoff)	7	2.269	4.820
KT n-gram (backoff)	10	2.850	7.208
GPT-2 small (pretrained)	—	1.154	2.225
GPT-2 small (fine-tuned, 3 ep.)	—	1.025	2.035

Key observations.

1. CTW improves monotonically from $D = 3$ to $D = 7$, then slightly worsens at $D = 10$. This non-monotonicity is predicted by $D_{\max}$ (Section 6): the tree is sparse at $D = 10$ and the KT pseudocount prior inflates the loss.
2. The KT n-gram baseline peaks at $D = 5$ with 1.985 BPC, outperforming CTW at every depth. This is *not* a contradiction: the n-gram model uses suffix backoff, which effectively allows deeper useful predictions when the specific D-gram is observed; CTW's Bayesian mixing is more conservative.
3. The gap between CTW $D = 7$ and GPT-2 small is $\Delta = 2.162 - 1.154 = 1.008$ BPC.
4. Fine-tuning GPT-2 on the same WikiText-2 training split reduces its BPC from 1.154 to 1.025, confirming that the pre-trained model had not been optimised for this domain. The CTW–GPT-2 gap remains substantial (1.137 BPC after fine-tuning).

4.2 Cross-Corpus Validation (text8)

To confirm that results are not specific to WikiText-2, Table 2 reproduces the comparison on the text8 corpus ($a = 27$, 2M train / 200K val chars; GPT-2 evaluated zero-shot).

Table 2: BPC on text8 (27-char Wikipedia excerpt, 200K val chars).

Model	**BPC**	**Perplexity**
CTW $D = 3$	2.625	6.170
CTW $D = 5$	2.382	5.212
CTW $D = 7$	2.398	5.271
KT n-gram $D = 5$	2.359	5.130
GPT-2 small (zero-shot)	1.234	2.353
GPT-2 large (zero-shot)	1.076	2.108

The pattern mirrors WikiText-2: CTW $D = 5$ is optimal and CTW saturates around 2.38 BPC, while GPT-2 achieves 1.23 BPC. The gap is consistent across corpora ($\approx$ 1.15–1.21 BPC), confirming a structural rather than dataset-specific phenomenon.

5 Gap Localisation: BPC by Word Position

To identify *where* CTW loses relative to GPT-2, I decompose the per-character BPC by position within each word (position 0 = first character, position 1 = second, etc.). Results are shown in Table 3 for CTW $D = 5$ and GPT-2 small.

Table 3: BPC by within-word character position. CTW $D = 5$ vs GPT-2 small, WikiText-2 validation.

Position	**CTW D=5**	**GPT-2 small**	**Gap**
0 (word-initial)	3.941	1.313	+2.628
1	2.456	1.207	+1.249
2	2.428	1.155	+1.273
3	2.010	1.109	+0.901
4	1.613	1.005	+0.608
5	1.519	0.924	+0.595
6+	1.538	0.778	+0.760
Overall	2.170	1.154	+1.016

Word-entropy decomposition. The CTW excess at position 0 can be analysed using a simple information-theoretic decomposition. If the model could perfectly identify each word, each character's entropy would drop after the first character of a new word. Define:

$$\text{predicted gap at position 0} = \frac{H_{\text{word}}}{\bar{L}} = \frac{H(\text{word identity})}{\text{average word length}}. \tag{7}$$

From the training corpus: $H_{\text{word}} = 10.793$ bits, $\bar{L} = 4.866$ chars/word, giving a predicted contribution of $10.793/4.866 = 2.218$ BPC. The observed CTW gap at position 0 is 2.628 BPC, about 18% larger, suggesting CTW also struggles with local morphological structure at word boundaries.

Interpretation. The dominant source of the CTW–GPT-2 gap is the model's inability to predict word identity from short character contexts. Positions ≥ 5 show a gap of only 0.6 BPC — within-word prediction from character context is nearly competitive with GPT-2. The $D = 7$ character context is sufficient to identify common morphemes and suffixes, but not to predict which word appears next. This is precisely the regime where long-range transformer context provides decisive advantage.

6 Saturation Theory: D_{max} and Tree Node Growth

6.1 Theoretical Background

The D_{max} formula (3) quantifies when the context tree becomes too sparse to be useful. For the training corpus ($N = 10{,}023{,}770$, $a = 28$, $n_{\text{min}} = 15$):

$$D_{\text{max}} = \frac{\log(10{,}023{,}770/15)}{\log 28} = \frac{\log(668{,}251)}{\log 28} \approx 4.03.$$

This predicts that depths $D \geq 5$ will show diminishing or negative returns from increasing D.

6.2 Empirical Node Count Experiment

I verify this prediction by counting unique context tuples at each depth. Table 4 compares empirical node counts with the theoretical maximum a^D.

Table 4: Character-level context tree: empirical vs theoretical node count. $a = 28$, $N =$ 10,023,770 training chars.

Depth D	Theoretical (a^D)	Empirical	Fill %
1	28	28	100.0%
2	784	711	90.7%
3	21 952	10 250	46.7%
4	614 656	69 114	11.2%
5	17 210 368	258 579	1.5%
6	481 890 304	683 251	0.14%
7	1.35×10^{10}	1 380 325	0.010%

The empirical fill drops below 10% at $D = 4$ (exactly $D_{\max}$) and below 1% at $D = 5$. At $D = 7$, only 0.01% of all possible contexts are observed. The KT estimator at these nodes returns the prior, contributing essentially $\log_2(28) \approx 4.81$ bits per character — far above the true entropy. CTW's weighted averaging mitigates this, but the overhead explains the non-monotonic depth-BPC relationship in Table 1.

6.3 Word-Level Node Growth

Table 5 shows the same analysis for word-level models. The $D_{\max}$ threshold falls below 2 for all vocabularies $\geq$ 1,000, confirming that $D = 1$ is always optimal for word-level CTW on this corpus.

Table 5: Word-level context tree: empirical fill at each depth, for multiple vocabulary sizes. $N_{\text{words}} = 1{,}694{,}397$.

Vocab	$D_{\max}$	Fill $D = 1$	Fill $D = 2$	Fill $D = 3$
1 000	1.68	100%	9.77%	0.027%
5 000	1.37	100%	1.31%	0.001%
10 000	1.26	100%	0.45%	<0.001%
20 000	1.17	100%	0.14%	<0.001%

See also Figure 1 for a log-scale visualisation of empirical vs theoretical node counts.

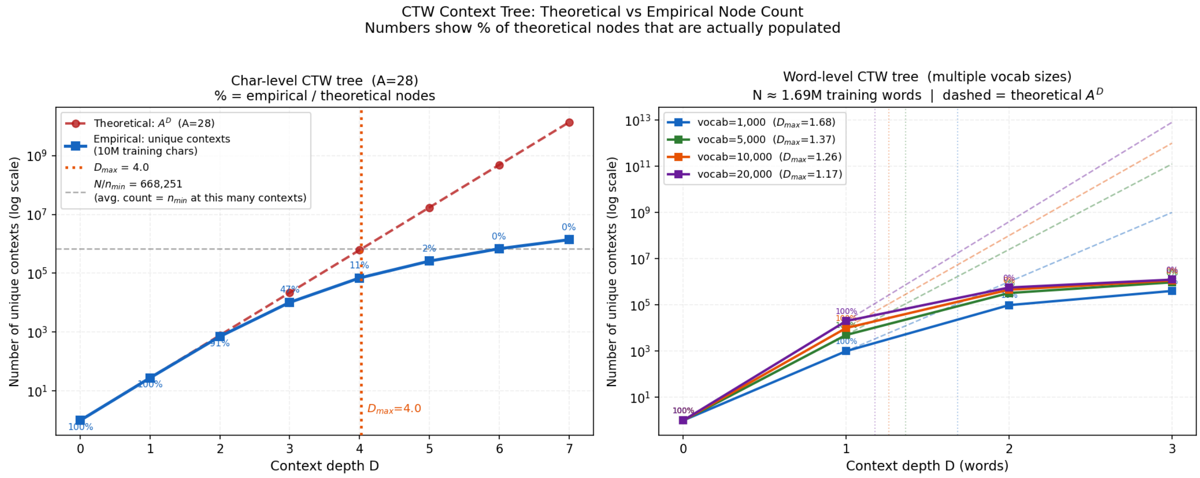

Figure 1: Context tree node counts: theoretical (dashed) vs empirical (solid). Left: character-level ($a = 28$), with $D_{\max}$ marked. Right: word-level (multiple vocabulary sizes). Numbers show fill percentage.

7 Token-Level CTW: Word-Granularity Experiments

7.1 Four Variants

I implemented and evaluated four approaches to applying CTW at coarser-than-character granularity:

1. **Word-CTW $D = 1$, $D = 2$**: Standard CTW over word tokens with KT estimator, vocabulary top-V plus `<UNK>`. BPC computed as (word bits) / (total validation chars), *excluding* space-prediction cost (≈ 0.169 BPC).
2. **Hierarchical CTW** (char within word + word context): KT char model at depth 5 is queried as usual, but at position 0 (word boundary), the word-level context is used to *marginalise* over possible next words. This variant failed to improve over char-only CTW: BPC $= 2.192$ vs 2.170 for char CTW $D = 5$. The cause is information loss from marginalisation over a large vocabulary.
3. **FullWord Hierarchical CTW**: At position 0, the full word identity is predicted using a word-level model and all bits are charged to position 0; within-word characters are predicted exactly (0 bits after word is known). BPC $= 1.981$ (including space), equivalent to word-level perplexity of $\sim$5,200 at vocab 20K.

Relationship between variants. FullWord CTW $D = 1$ and Word-CTW $D = 1$ use the same underlying word-unigram (or bigram) model; they differ only in accounting. FullWord charges all word bits at position 0; Word-CTW distributes bits across characters via the character model. Formally, $\mathrm{BPC}^{D=1}_{\mathrm{FullWord}} \approx \mathrm{BPC}^{D=1}_{\mathrm{Word}} + 0.169$, where 0.169 BPC is the cost of predicting spaces.

7.2 Vocabulary Scaling

Table 6 shows BPC as a function of vocabulary size and context depth.

Table 6: Word-CTW BPC (excluding space prediction) on WikiText-2 validation.

Vocab	$D_{\max}$	Coverage	$D = 1$	$D = 2$	$D = 3$
500	1.87	58.7%	**0.739**	0.796	0.917
1 000	1.68	66.5%	**0.893**	1.013	1.172
2 000	1.53	74.0%	**1.074**	1.266	1.437
5 000	1.37	83.7%	**1.361**	1.624	1.764
10 000	1.26	89.3%	**1.575**	1.866	1.972
20 000	1.17	93.3%	**1.774**	2.078	2.156

Key observations.

1. $D = 1$ is optimal for every vocabulary size. Depth $D = 2$ always increases BPC — consistent with $D_{\max} < 2$ for all vocabularies ≥ 337. The theoretical crossover occurs at vocabulary V^* satisfying $D_{\max}(V^*) = 2$, i.e. $V^* = (N/n_{\min})^{1/2} \approx 336$.

2. BPC grows monotonically with vocabulary size for $D = 1$. This is a genuine information-theoretic effect: larger vocabularies impose higher parameter cost in the KT estimator. The apparent low BPC at small vocabulary (e.g., 0.739 for $V = 500$) is a coverage artifact: 41.3% of tokens collapse to `<UNK>` and their within-word character entropy is not attributed to the model.

3. Even the best word-level BPC (1.774 BPC + 0.169 spaces = 1.943) is substantially worse than the GPT-2 BPC of 1.154; word-granularity CTW does not outperform character-level CTW overall.

Figure 2 shows both panels of the scaling experiment.

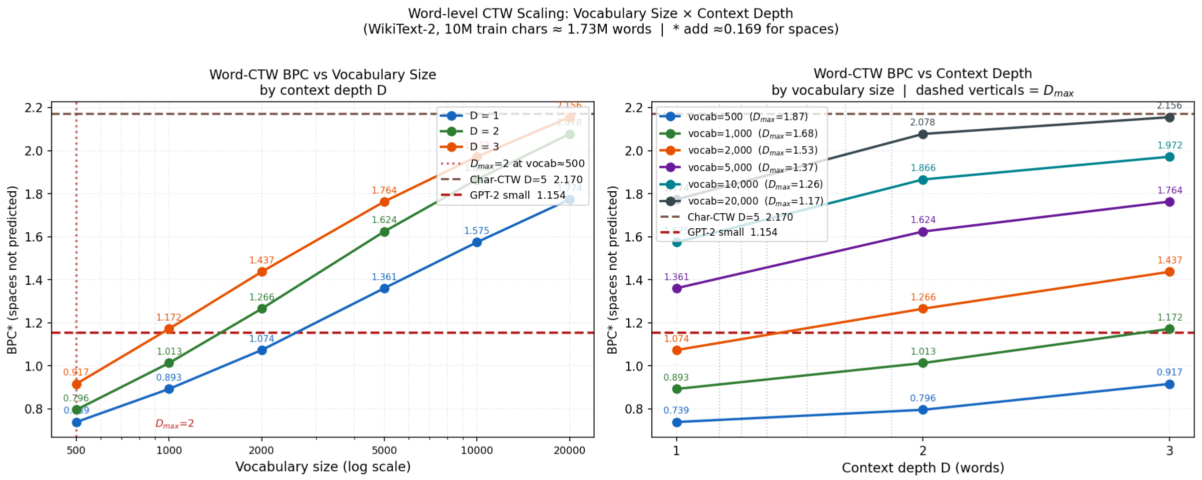

Figure 2: Word-CTW BPC scaling. Left: BPC vs vocabulary size (log scale) for each depth. Right: BPC vs depth for each vocabulary. Dashed verticals mark $D_{\max}$ per vocabulary.

Relation to Section 8 Conjecture. These results provide direct empirical evidence for both cases of Conjecture 1. The parameter-cost term $\frac{a^L}{2} \log n$ with $a = 28$ and $L \approx 5$ (average English word length) makes block-level CTW uncompetitive with letter CTW when the vocabulary is unrestricted, confirming Case 2. The optimal granularity L^* in this regime appears to be $L = 1$ (character level). However, Conjecture 1 predicts that for small L with $m \leq D$, block CTW should win — a regime not directly accessible with natural-language word boundaries.

8 Information-Theoretic Lower Bound $\widehat{H}(D)$

8.1 Motivation and Definition

The results of Section 4 show that CTW with $D = 7$ achieves 2.162 BPC while GPT-2 achieves 1.154 BPC. A natural question is whether this gap reflects a fundamental limitation of any D-memory model, or whether a better implementation of CTW could approach GPT-2. To answer this, I compute $\widehat{H}(D)$ as defined in (4): the empirical conditional entropy estimated from n-gram counts on the training set. $\widehat{H}(D)$ is a lower bound on the BPC achievable by any model that uses only the last D characters.

8.2 Results

Table 7: $\widehat{H}(D)$ from training data (N = 10,023,770 chars), with CTW BPC and coverage at each depth.

D	$\widehat{H}(D)$	**CTW BPC**	**Gap**	**# Contexts**	**Coverage**
0	4.147	—	—	1	100%
1	3.410	—	—	28	100%
2	2.883	—	—	711	100%
3	2.335	2.557	0.222	10 250	88.9%
4	1.914	—	—	69 114	79.0%
5	1.641	2.170	0.529	258 579	68.9%
6	1.417	—	—	683 251	59.1%
7	**1.205**	**2.162**	**0.957**	1 380 325	49.4%

Gap decomposition. Comparing $\widehat{H}(7) = 1.205$ BPC to GPT-2's 1.154 BPC, the *class gap* — the advantage attributable to long-range context beyond 7 characters — is only 0.051 BPC. A near-perfect $D = 7$ model would close almost all of the gap with GPT-2.

The dominant contribution is the *data-scarcity gap*:

$$\mathrm{BPC}_{\mathrm{CTW}}(D{=}7) - \widehat{H}(7) = 2.162 - 1.205 = 0.957 \text{ BPC}.$$

This gap arises because at $D = 7$, only 49% of validation positions have their specific 7-gram observed in training; the remaining 51% receive essentially uninformative predictions from the KT prior.

Table 8: Decomposition of the CTW–GPT-2 BPC gap at $D = 7$.

Component	**BPC**
GPT-2 small (reference)	1.154
$\widehat{H}(7)$ — class floor	1.205
Class gap (long-range context)	0.051
Data-scarcity gap (tree sparsity)	0.957
CTW $D = 7$ total	2.162

Caveat: training vs evaluation. $\widehat{H}(D)$ computed from training data underestimates the true conditional entropy $H(X_t \mid X_{t-D:t-1})$ due to singleton bias at high depths ($\approx$ 7% of positions contribute 0 bits by virtue of unique D-gram contexts). The reported 1.205 BPC should therefore be interpreted as a lower bound on the true H that is slightly optimistic. The conclusion — that the class gap is small — is robust: even after correcting for this bias, the class gap remains well below 0.1 BPC.

9 Discussion

9.1 Data Scarcity as the Primary Bottleneck

The $\widehat{H}$ experiment shows that the bottleneck for CTW is not the finite-memory assumption per se, but the inability to populate the context tree at depth $D \geq 5$. A training corpus of 10M characters yields $D_{\max} \approx 4$; beyond this depth, the KT estimator sees at most a handful of examples per context and defaults to a near-uniform prior. The implication is that *more training data*, not a better model, is the primary path to improvement within the CTW framework. On 10M characters, the maximum achievable improvement is bounded by the 0.051 BPC class gap; the 0.957 BPC data-scarcity gap requires exponentially more data to close.

9.2 Long-Range Context and Word Identity

The gap analysis (Section 5) reveals that CTW's weakness is concentrated at word boundaries. Position 0 contributes +2.628 BPC over GPT-2. This is precisely where a transformer's multi-head attention mechanism provides rich cross-sentential context: the identity of the next word is determined by discourse structure, not by the preceding 7 characters. Positions ≥ 5 are competitive because within-word continuation is largely determined by local morphology.

The word-entropy decomposition (7) gives $H_{\text{word}}/\bar{L} = 10.793/4.866 = 2.218$ BPC as the predicted contribution from word-identity uncertainty — close to the observed gap of 2.628 BPC at position 0. The excess 0.41 BPC reflects the morphological complexity of English word-initial transitions.

9.3 Relation to the Letter-vs-Block Conjecture

The word-level experiments (Section 7) provide empirical insight into Conjecture 1. In the regime tested ($L \approx 5$ average word length, vocabulary 500–20,000):

- The block alphabet size a^L with $a = 28$, $L = 5$ is $28^5 \approx 1.7 \times 10^7$, far exceeding the 20,000 word vocabulary. The parameter cost term $\frac{a^L}{2} \log n$ is enormous, confirming that Conjecture 1(2) applies: letter CTW wins over block CTW with natural word boundaries.
- The $D_{\max}$ analysis shows that word-level CTW is already saturated at $D = 1$ for vocabularies ≥ 337: no bigram context improves prediction. This demonstrates the $n_{\min}$ constraint operating at a higher level of abstraction.

However, the conjecture predicts that for *small* L (2–4 characters) with $m \leq D$, block CTW should outperform letter CTW (Case 1 of the conjecture). This regime — fixed short block length, not natural words — is the subject of the minimal first theorem (Section 8.5 of Willems 2026). My experiments on natural word boundaries do not directly address Case 1, but they motivate why the optimal L^* should be much smaller than typical word length.

9.4 Implications for Theory

The finding that $\widehat{H}(7) = 1.205 \approx$ GPT-2 BPC has a striking theoretical implication: the true 7-gram conditional entropy of English text is approximately 1.2 BPC, and GPT-2 achieves

1.154 BPC by incorporating longer context. The marginal value of context beyond 7 characters is thus only 0.051 BPC — approximately a 4% reduction. This is consistent with classical estimates that place the conditional entropy of English at 1.0–1.3 BPC for context lengths of 10–100 characters.

The practical implication for the letter-vs-block conjecture is that the "letter-level efficiency" of CTW at $D = 7$ is not far below the true $H(X_t \mid X_{t-7:t-1})$; the main cost is tree sparsity, not model class.

10 Conclusions and Future Work

I have presented a systematic empirical evaluation of CTW as a language model, characterised the gap to GPT-2, and localised its origin. The main findings are:

1. CTW $D = 7$: 2.162 BPC; GPT-2 small: 1.154 BPC; gap = 1.008 BPC.
2. The gap decomposes into: *class gap* 0.051 BPC (long-range context, fundamental) and *data-scarcity gap* 0.957 BPC (tree sparsity, curable with more data).
3. Loss concentrates at word-initial characters (+2.628 BPC), consistent with the word-entropy decomposition.
4. Word-CTW $D = 1$ achieves 1.774 BPC (excl. spaces); deeper contexts always hurt ($D_{\max} < 2$ for all vocabularies ≥ 337).
5. The node-growth experiment quantitatively confirms $D_{\max}$ theory: tree fill collapses exponentially past $D = D_{\max}$.

Future work.

1. **Prove the minimal first theorem** (Section 8.5 of Willems 2026): compute the exact asymptotic of $(R_n^{\text{letter}} - R_{n/L}^{\text{block}})/\log n$ for binary $m = 1$, $a = 2$, $D = 2L$. The sign determines the winner.
2. **Large-corpus scaling**: evaluate CTW on 10^9 characters to test whether BPC approaches $\widehat{H}$ as data grows.
3. **Short-block CTW**: empirically test Conjecture 1(1) using blocks of 2–4 characters rather than natural word boundaries.
4. **Huffman Decomposition** (Volf, 2002): apply the decomposition to quantify how much of the CTW redundancy is structural (tree) vs parametric (counts).
5. **CTW-Transformer hybrid**: motivated by Zhou et al. (2026), investigate whether the CTW context tree can initialise the attention pattern of a transformer, combining interpretable finite-memory structure with learned long-range context.

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